Diamondx Tester
Air-Cooled Universal Test Platform

Overview

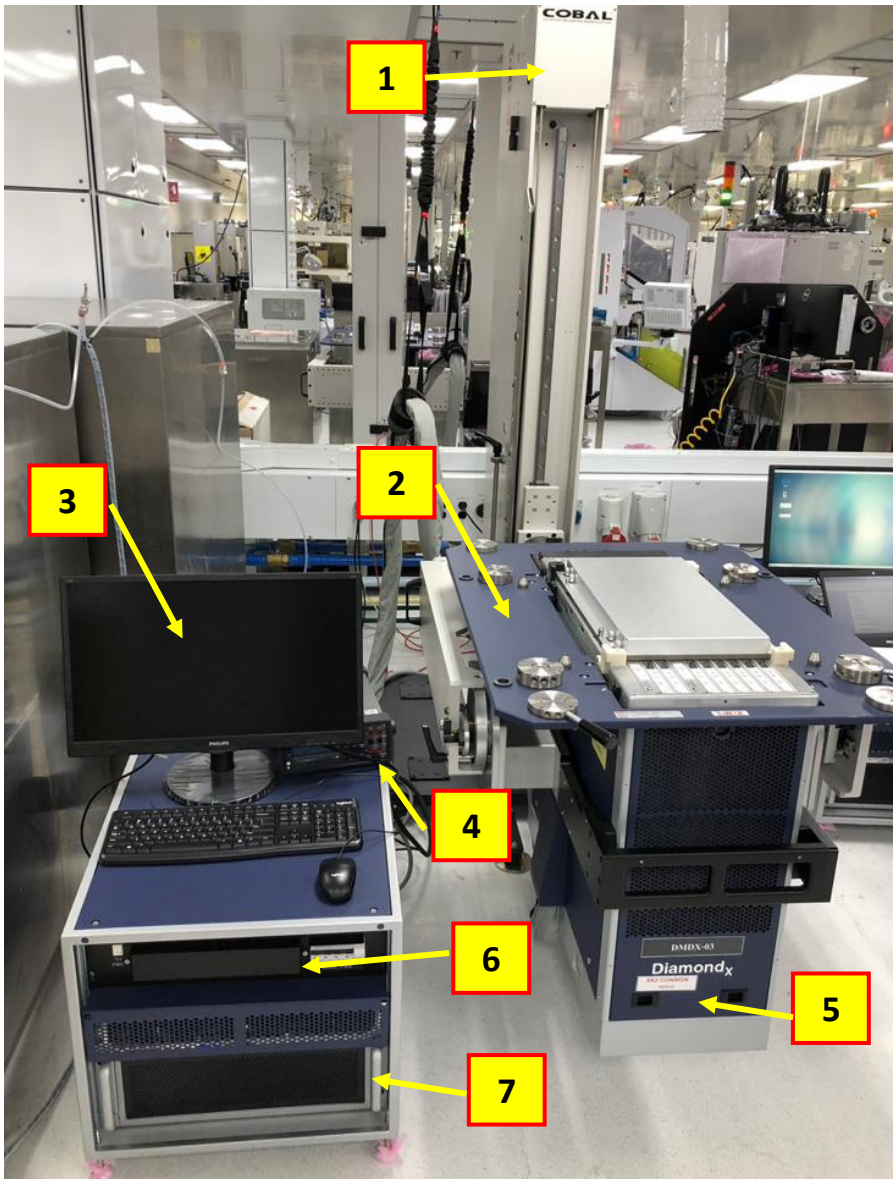
- Linux CentOS7
- Electrical requirement 190-240VAC (208VAC NORMINAL) 50/60HZ, 3PHASE 15A

Power Supply

Power inside the test system is distributed as 48V DC, and as separate outlet strips for the computer, monitor, and other uses.

48VDC Power

The DiamondX Test System has a main 48V DC power supply within the system resource box. The power supply is separated into three bricks that supply power to the testhead. The power supplies are over-current protected and cooled by forced air.



1. Manipulator
2. Tester Docking Plate
3. Monitor
4. DMM34461A
5. Testhead
6. Individual Power Sections (Under this cover)
7. PC Computer
8. Ejector

NOTE: Item6 and 7 is under **extended system resource box**

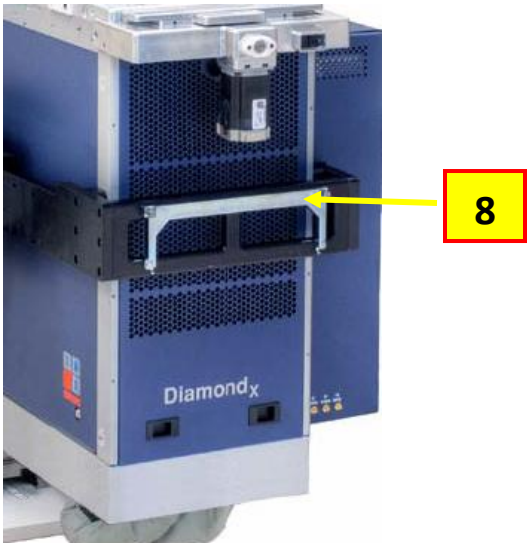


Figure 11. Power Distribution Unit, rear view

The outlets marked Utility are used for the system DMM, computer, monitor, and fans. These outlets are not turned off by the Testhead Power switch, they are only turned off by the Main Power circuit breaker.

Contents

1. Tester Power Up
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7. การอ่านชื่อบอร์ด, P/N และ S/N
8. การถอด-ใส่ System Boards ในเทสเฮดท์
9. การใส่-ถอด Load board

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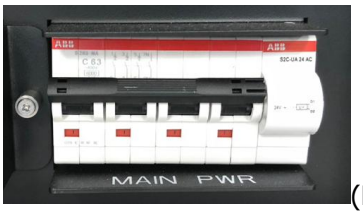
1. Tester Power Up

1.1 Turn-On 'MAIN PWR'



(Main Power of tester)

Note:

Username: Service

Password: -

1.2 Turn-On 'TH PWR'

(Power Supply)

1.3 Turn-On PC Computer switch

(HDD Monitor LED)

***NOTE:* Main Power Circuit Breaker**

The Main Power circuit breaker powers on or off all equipment in the tester, including the computer and monitor.

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2. Tester Power Down

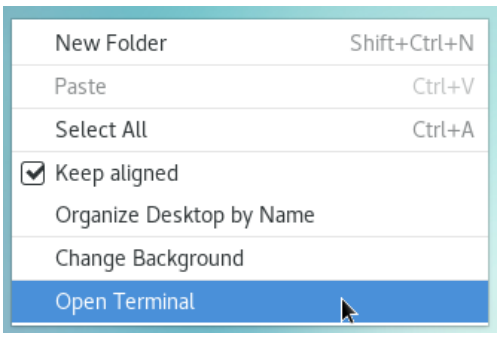
2.1 กดปุ่ม Shutdown ด้านขวาบนมุมจอของคอมพิวเตอร์

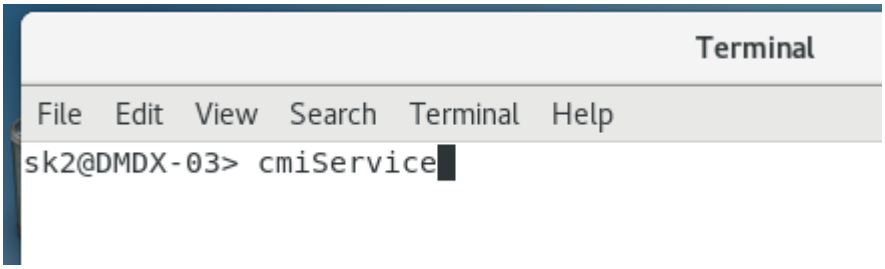
2.2 Turn-Off 'TH PWR'

2.3 Turn-Off 'MAIN PWR'

3. Checkers + Calibrations + Verifications (Load boards Remove)

3.1 คลิกขวาที่ว่างที่หน้าจอแล้วเลือก Open Terminal แล้วพิมพ์ cmiService แล้วกด Enter (เป็นการเช็คdata log path, system detected board, Configของเทสเตอร์ ส่วนใหญ่แล้วจะทำหลังจากเปิดเครื่องหรือรีสตาร์ทคอมพิวเตอร์ และห้ามไปขัดจังหวะใดๆระหว่างทำ cmiService)





Example: cmiService

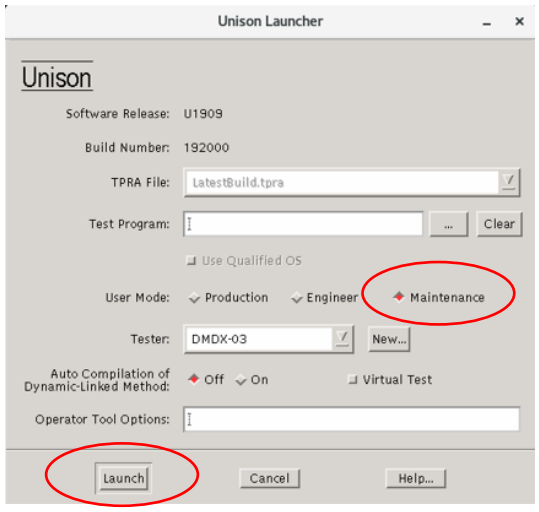
```
sk2@DMDX-05> cmiService
cmiService: Opening logFile: /opt/ltx_nic/log/cmiService.28.cmilog
 Step1: BusDetect
 Step2: Instantiate instruments. Unison BuildId: 194200, ConfigFile: /opt/ltx_nic/pwrup/unison_config
  Created Instrument:IMA0_Slot30 Slot[30]:LTXC_PHX_THCTL, PN# 979-1297, SN# 11594830, VRev 10.NE
  Created Instrument:IMA0_Slot0 Slot[0]:LTXC_PHX_SDU, PN# 979-1300, SN# 11595154, VRev 10.BA
  Created Instrument:IMA0_Slot6 Slot[6]:LTXC_PHX_GX1_256, PN# 979-1597, SN# 11549504, VRev 21.AB
  Created Instrument:IMA0_Slot7 Slot[7]:LTXC_PHX_GX1_256, PN# 979-1597, SN# 11550005, VRev 21.AD
  Created Instrument:IMA0_Slot17 Slot[17]:LTXC_PHX_HDVI, PN# 982-7642, SN# 2218022, VRev 00.P
 Step3: InstrumentTable config
 Step3B: PowerOnReset
 Step4: Probe instruments, SystemType: Diamondx
   Probing: IMA0_Slot0 Slot[0]:LTXC_PHX_SDU, PN# 979-1300, SN# 11595154, VRev 10.BA
   Probing: IMA0_Slot6 Slot[6]:LTXC_PHX_GX1_256, PN# 979-1597, SN# 11549504, VRev 21.AB
   Probing: IMA0_Slot7 Slot[7]:LTXC_PHX_GX1_256, PN# 979-1597, SN# 11550005, VRev 21.AD
   Probing: IMA0_Slot17 Slot[17]:LTXC_PHX_HDVI, PN# 982-7642, SN# 2218022, VRev 00.P
   Probing: IMA0_Slot30 Slot[30]:LTXC_PHX_THCTL, PN# 979-1297, SN# 11594830, VRev 10.NE
   Probing: IMA0_ImaBackplane Slot[31]:LTXC_ImaBackplane, PN# 974-1293, SN# 11594802, VRev 00.CA
   Probing: LBRD_ITEM0 Slot[240]:LTXC_LoadBoardHandler
 Step5: Configure ResourceManager
 Step6: Configure instruments
   Config:  IMA0_ImaBackplane Slot[31]:LTXC_ImaBackplane, PN# 974-1293, SN# 11594802, VRev 00.CA
   Config:  LBRD_ITEM0 Slot[240]:LTXC_LoadBoardHandler
   Config:  IMA0_Slot6 Slot[6]:LTXC_PHX_GX1_256, PN# 979-1597, SN# 11549504, VRev 21.AB
   Config:  IMA0_Slot7 Slot[7]:LTXC_PHX_GX1_256, PN# 979-1597, SN# 11550005, VRev 21.AD
   Config:  IMA0_Slot17 Slot[17]:LTXC_PHX_HDVI, PN# 982-7642, SN# 2218022, VRev 00.P
   Config:  IMA0_Slot0 Slot[0]:LTXC_PHX_SDU, PN# 979-1300, SN# 11595154, VRev 10.BA
   Config:  IMA0_Slot30 Slot[30]:LTXC_PHX_THCTL, PN# 979-1297, SN# 11594830, VRev 10.NE, UniqId 400016e30c000affff2a001333fcc6bc
 Step7: Initialize system
     Slot[240]:No Loadboard : NotPresent
  IMA0-EventAlign rtc checked for 5 slots. MRtc=e4, EvRtc=111, DsRTC=8e
 CmiCapConnection registered.

CMI service ready: Fri Oct  7 10:29:27 2022 :: /opt/ltx/releases/U1909/updates/194200/x86_64_linux_3.10.0
// Skipping already loaded FPGA: CPIn Rev: 1.4.12 (0x01040c) in PHX_THCTL:IMA0_Slot30
// Skipping already loaded FPGA: CPIn Rev: 1.4.12 (0x01040c) in PHX_SDU:IMA0_Slot0
// Skipping already loaded FPGA: CPIq Rev: 1.1.29 (0x01011d) in PHX_GX1_256:IMA0_Slot6
// Skipping already loaded FPGA: CPIq Rev: 1.1.29 (0x01011d) in PHX_GX1_256:IMA0_Slot7
// Skipping already loaded FPGA: CPId Rev: 1.2.37 (0x010225) in PHX_HDVI:IMA0_Slot17
// Skipping already loaded FPGA: EVS Rev: 1.2.16 (0x010210) in PHX_THCTL:IMA0_Slot30
// Skipping already loaded FPGA: THC Rev: 1.3.9 (0x010309) in PHX_THCTL:IMA0_Slot30
// Skipping already loaded FPGA: THCFlash Rev: 0.0.0 (0x010309) in PHX_THCTL:IMA0_Slot30
// Skipping already loaded FPGA: SDU Rev: 1.1.3 (0x010103) in PHX_SDU:IMA0_Slot0
// Skipping already loaded FPGA: STG Rev: 1.4.123 (0x01047b) in PHX_GX1_256:IMA0_Slot6
// Skipping already loaded FPGA: STG Rev: 1.4.123 (0x01047b) in PHX_GX1_256:IMA0_Slot7
// Skipping already loaded FPGA: STG Rev: 1.4.123 (0x01047b) in PHX_GX1_256:IMA0_Slot6
// Skipping already loaded FPGA: STG Rev: 1.4.123 (0x01047b) in PHX_GX1_256:IMA0_Slot7
// Skipping already loaded FPGA: STG Rev: 1.4.123 (0x01047b) in PHX_GX1_256:IMA0_Slot6
// Skipping already loaded FPGA: STG Rev: 1.4.123 (0x01047b) in PHX_GX1_256:IMA0_Slot7
// Skipping already loaded FPGA: HDVI_BI Rev: 7.0.5 (0x070005) in PHX_HDVI:IMA0_Slot17
Checking if FLASH is blank.
// Loading HDVI_IC, Slot 17, Rsrcs 800 with:: /opt/ltx/releases/U1909/updates/194200/x86_64_linux_3.10.0/plugins/config/optional/PHX/hdvi/1.22.105/fpga/hdvi_ic_5_03_6_wc.svf
FPGA load done, performing system re-initialization.
  IMA0-EventAlign rtc checked for 5 slots. MRtc= c, EvRtc=101, DsRTC=79
Closing CmiDriver
sk2@DMDX-05>
```

3.2 เช็คก่อนว่า DMM34461A นั้นมี GPIB cable ต่ออยู่และตัวเครื่องต้องเปิดอยู่ (Power-On) //ถ้าไม่ต่อDMM34461A จะทำให้ initial ไม่ผ่าน

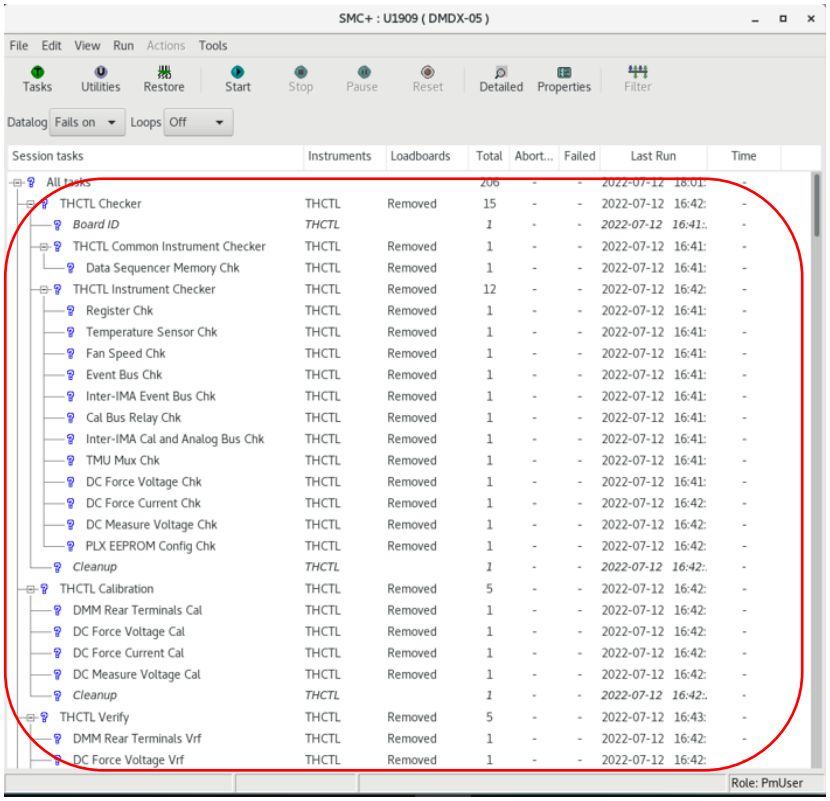
34461A Multimeter

The Keysight 34461A is a 6.5-digit multimeter (Cohu Part 118-8375-00) used to perform system calibrations and checks. The multimeter is connected to the computer GPIB. The GPIB is set to address "22", while the multimeter's ID is set to "HP34401A". The power switch for the meter is on the front of the meter.

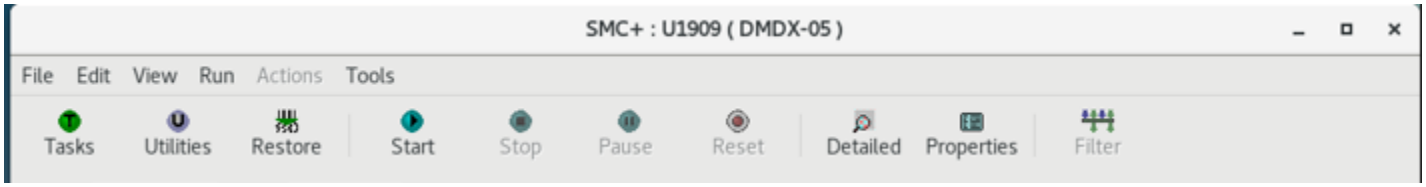
3.3 พิมพ์ launcher & แล้วกด Enter จะมีหน้าต่างด้านล่างนี้ขึ้นมา ให้เลือก Maintence แล้วคลิก Launch //คำสั่งเรียก Unison Launcher



3.4 รอจนกว่า Initiating Unison sequence เสร็จ



3.**5 เลือก File --> Change Role --> *SysUser* --> Task**

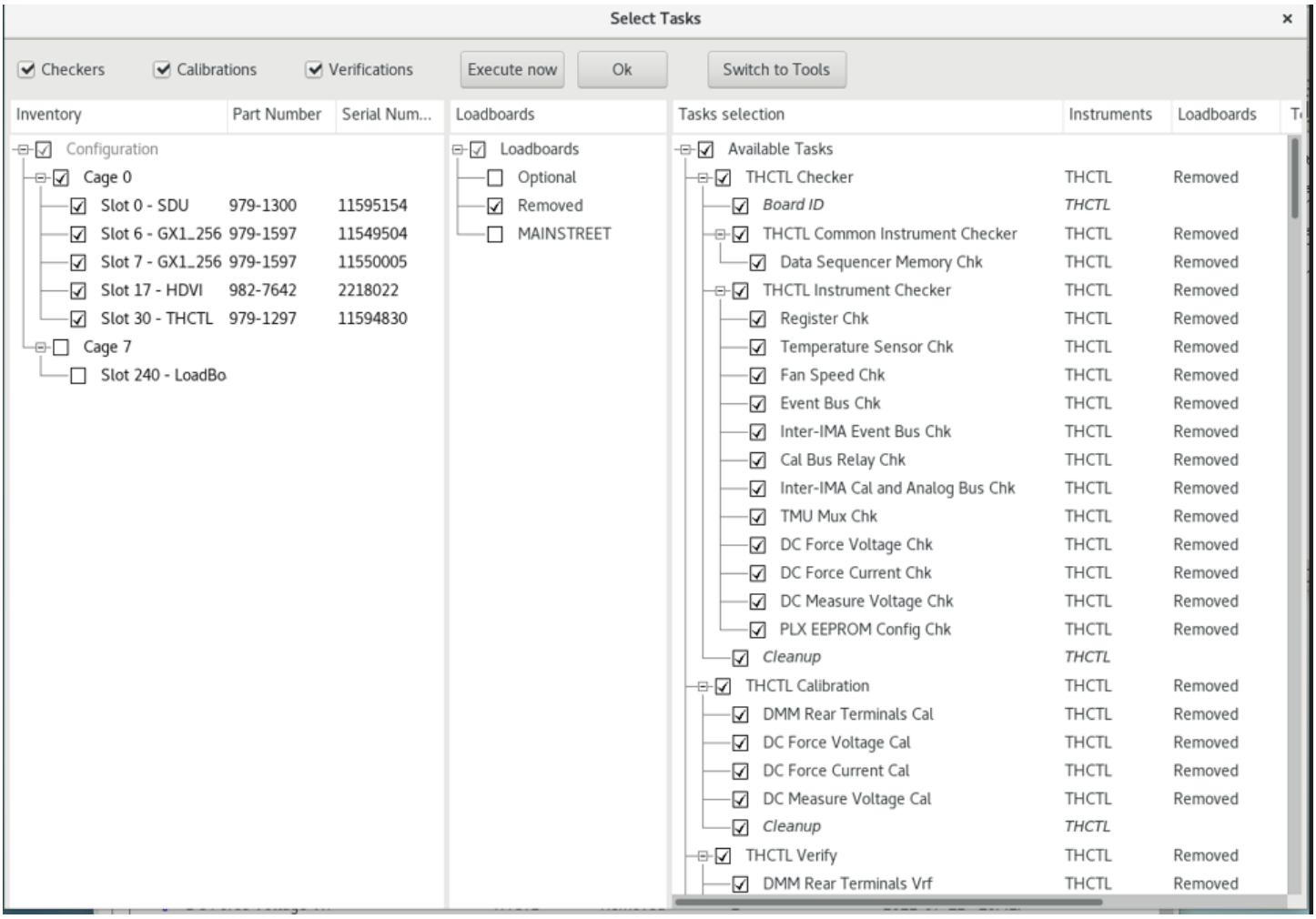


Inventory: ให้เลือกบอร์ดที่ต้องการ

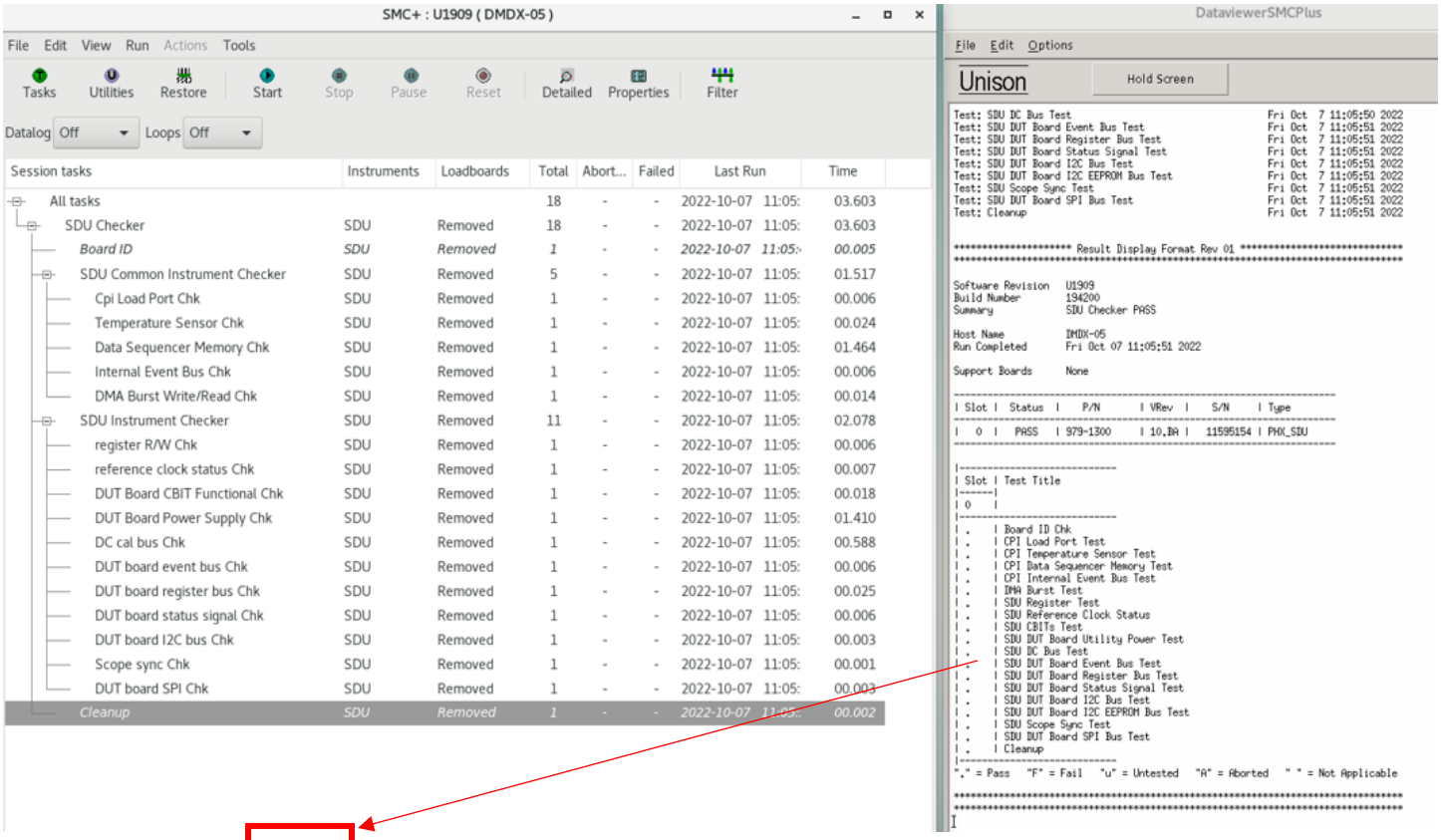
Loadboards: ให้เลือก Remove for with out Cal board

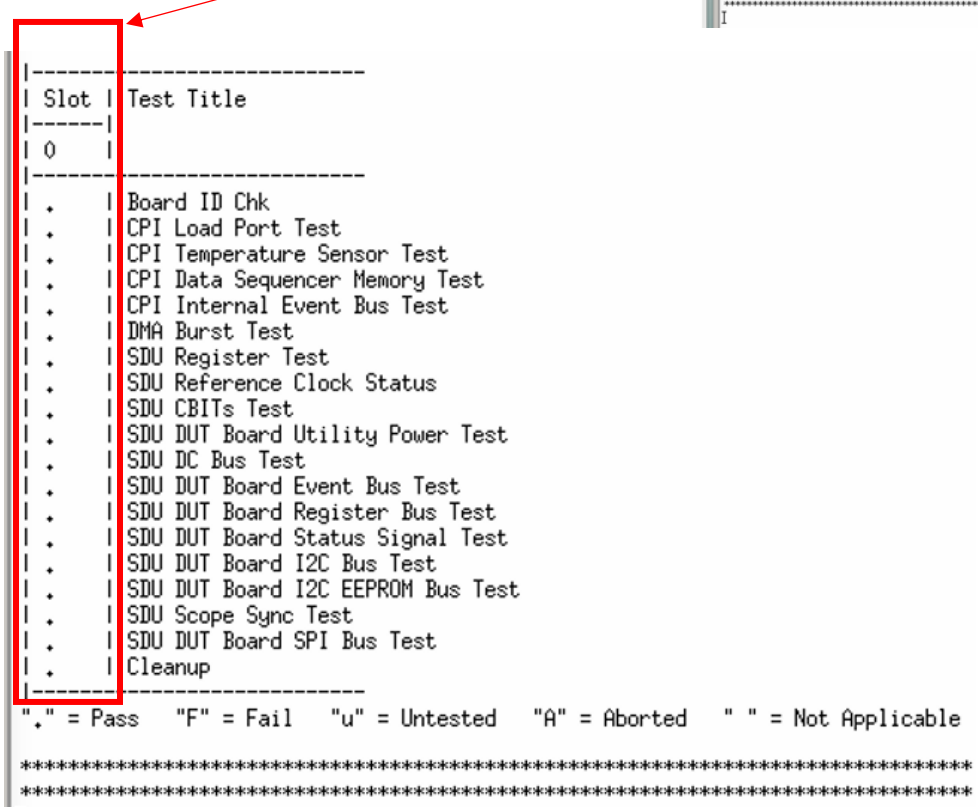
Task selection: ให้เลือกบอร์ดที่ต้องการ

จากนั้นกด "Execute now"



3.6 การดูผลPass/Fail ถ้ามีเครื่องหมาย (.) ตรงคอลัมป์สล๊อท แสดงว่าบอร์ดผ่านหมด





หรือดู Summary คอลัมม์Failต้องเป็น0

Version: U1909
Tester: DMDX-04
Created: 2022-10-07 09:39:06

Session tasks	Instruments	Loadboards	Total	Aborted	Failed	Last Run	Duration (sec.)
All tasks			29	0	0	2022-10-07 09:39:06	264.200
SDU Checker	SDU	MAINSTREET	3	0	0	2022-10-07 09:34:54	12.790
Board ID	SDU	MAINSTREET	1	0	0	2022-10-07 09:34:41	00.008
SDU Instrument Checker	SDU	MAINSTREET	1	0	0	2022-10-07 09:34:54	12.775
DUT IO Chk	SDU	MAINSTREET	1	0	0	2022-10-07 09:34:54	12.775
Cleanup	SDU	MAINSTREET	1	0	0	2022-10-07 09:34:54	00.007
GX1 Checker	GX1_256	MAINSTREET	7	0	0	2022-10-07 09:36:20	85.589
Disable DUT safe mode	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:34:54	00.018
Board ID Chk	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:34:54	00.010
Stabilize Temperature	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:36:18	84.211
LoadDefaultCal	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:36:19	00.715
GX1 Instrument Checker	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:36:20	00.623
DUT IO Chk	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:36:20	00.623
Enable DUT safe mode	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:36:20	00.007
Clean Up	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:36:20	00.003
GX1 Calibration	GX1_256	MAINSTREET	9	0	0	2022-10-07 09:37:50	90.260
Disable DUT safe mode	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:36:20	00.012
LoadCal	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:36:21	00.973
Init DAC Control	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:36:21	00.002
Stabilize Temperature	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:36:42	21.060
Timing Calibration	GX1_256	MAINSTREET	2	0	0	2022-10-07 09:37:50	68.002
Drive Alignment Cal	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:37:45	62.618
Compare Alignment Cal	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:37:50	05.384
SaveCal	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:37:50	00.197
Enable DUT safe mode	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:37:50	00.007
Clean Up	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:37:50	00.003
GX1 Verify	GX1_256	MAINSTREET	5	0	0	2022-10-07 09:38:32	41.985
Disable DUT safe mode	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:37:50	00.013
LoadCal	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:37:51	00.971
Stabilize Temperature	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:38:12	21.061
Timing Verify	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:38:32	19.936
Grp8 Timing Accuracy Vrf	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:38:32	19.936
Enable DUT safe mode	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:38:32	00.002
HDVI Checker	HDVI	MAINSTREET	3	0	0	2022-10-07 09:38:52	19.334
Record Temperature for Chk	HDVI	MAINSTREET	1	0	0	2022-10-07 09:38:34	01.282
DUT I/O Chk	HDVI	MAINSTREET	1	0	0	2022-10-07 09:38:50	16.944
Cleanup	HDVI	MAINSTREET	1	0	0	2022-10-07 09:38:52	01.107
Phx Digital Slot-to-Slot Timing CAL	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:38:59	07.406
Phx Digital Slot Cal	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:38:59	07.406
Phx Digital Slot-to-Slot Timing Verify	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:39:06	06.833
Phx Digital Slot Vrf	GX1_256	MAINSTREET	1	0	0	2022-10-07 09:39:06	06.833

4. Checkers + Calibrations + Verifications (Main Street)

ทำเหมือนข้อ A. ทุกประการ แต่เปลี่ยนตรง Loadboards

Loadboards: ให้เลือก MAINSTREET

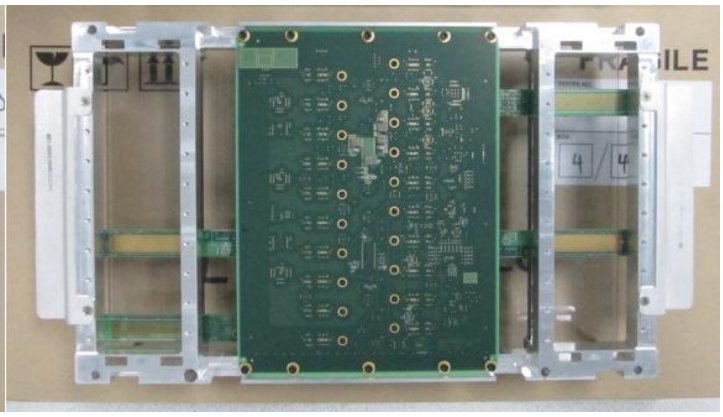
ปล. ต้องใส่บอร์ด MainStreet ก่อนเป็นอันดับแรก "**CAL-MAIN-DMDX/A-01**"

Note: CAL-MAIN-DMDX/A-01 (SK2 config)

CAL-MAIN-DMDX/A-02 (STM config)

CAL-MAIN-DMDX/A-03 (STM config)

CAL-MAIN-DMDX/A-04 (SS0 config)

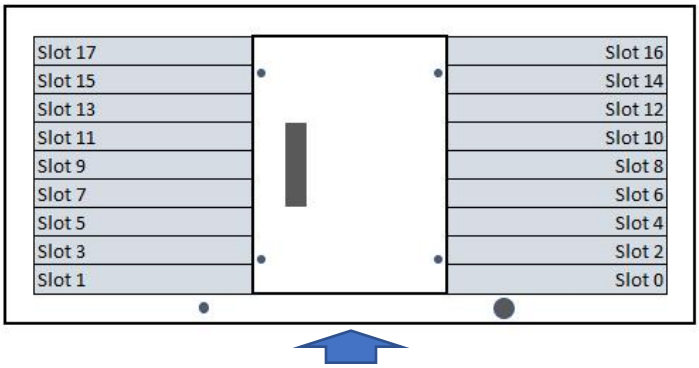


MAINSTREET BOARD

6. Existing DiamondX tester configurations

MC Owner	UTL	Consigned	Consigned	Consigned	Consigned	Consigned	Consigned	UTL	UTL	Comment
Tester No	DMDX-01	DMDX-02	DMDX-03	DMDX-04	DMDX-05	DMDX-06	DMDX-07	DMDX-08	DMDX-10	
Tester S/N	90858	90225	91010	91133	91134	91135	91136	91016	90467	
Config	STM	Sciosense	SK2	SK2	SK2	SK2	SK2	STM	STM	
Slot										
0	SDU	SDU	SDU	SDU	SDU	SDU	SDU	SDU	SDU	*Fix Slot*
1	-	HDVI	-	-	-	-	-	-	QTMPx	
2	-	HDVI	-	-	-	-	-	-	QFVIx	
3	-		-	-	-	-	-	-	QFVIx	
4	-	DPIN96	-	-	-	-	-	-	HVVIx	
5	-		-	-	-	-	-	-	DCTMx	
6	-	HDVI	GX1x-1	GX1x-1	GX1x-1	GX1x-1	GX1x-1	-		*Fix Slot*
7	-	HDVI	GX1x-2	GX1x-2	GX1x-2	GX1x-2	GX1x-2	-		*Fix Slot*
8	GX1x-1	DPIN96	-	-	-	-	-	GX1x-1	GX1	*Fix Slot*
9	GX1x-2	DPIN96	-	-	-	-	-	GX1x-2	GX1	*Fix Slot*
10	PMVIx-1	HDVI	-	-	-	-	-	PMVIx-1	PMVIx	
11	PMVIx-2		-	-	-	-	-	PMVIx-2	PMVIx	
12	-	DPIN96	-	-	-	-	-	-	QFVIx	
13	-	DPIN96	-	-	-	-	-	-	QFVIx	
14	PMVIx-3	DPIN96	-	-	-	-	-	PMVIx-3	PMVIx	
15	-	DPIN96	-	-	-	-	-	-	PMVIx	
16	-		-	-	-	-	-	-	QFVIx	
17	-	HDVI	HDVI	HDVI	HDVI	HDVI	HDVI	-	HVVIx	*Fix Slot*
18	-		-	-	-	-	-	-	DCTMx	
19	-		-	-	-	-	-	-	QFVIx	
20	-		-	-	-	-	-	-		
30	THCTL	THCTL	THCTL	THCTL	THCTL	THCTL	THCTL	THCTL	THCTL	*Fix Slot*
OS	Centos7.2	Centos7.2	Centos7.6	Centos7.6	Centos7.6	Centos7.6	Centos7.6	Centos7.6		

7. การนับ Board Slot Number



Testhead Front

8. การอ่านชื่อบอร์ด, P/N และ S/N

จากรูปบอร์ดตัวอย่าง

Board Name: HDVI

P/N: 982-7642-00

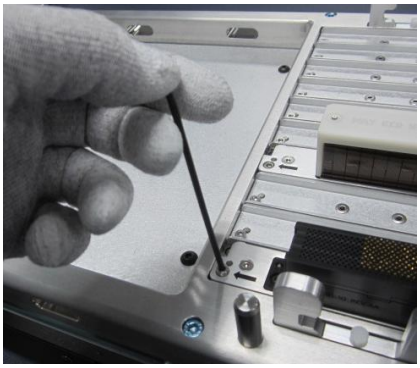
S/N: 2212084

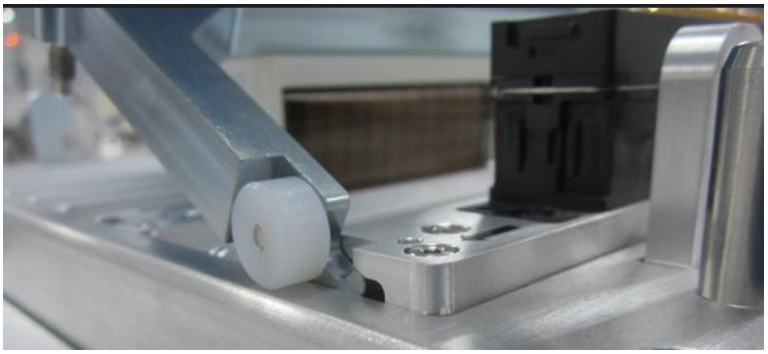
REV: OB

9. การถอด-ใส่ System Boards ในเทสเฮดท์

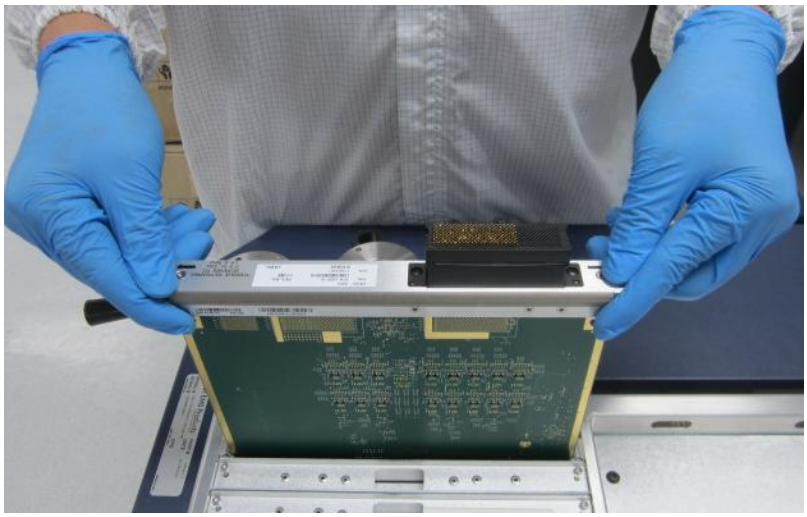
9.1 (การถอด)Turn-Off 'TH PWR'

9.2 จากรูปให้คลายสกรูทั้ง2ตัว(ที่มีลูกศรชี้) แล้วงัดบอร์ดขึ้นมาด้วย Ejector แล้วดึงบอร์ดขึ้นมา

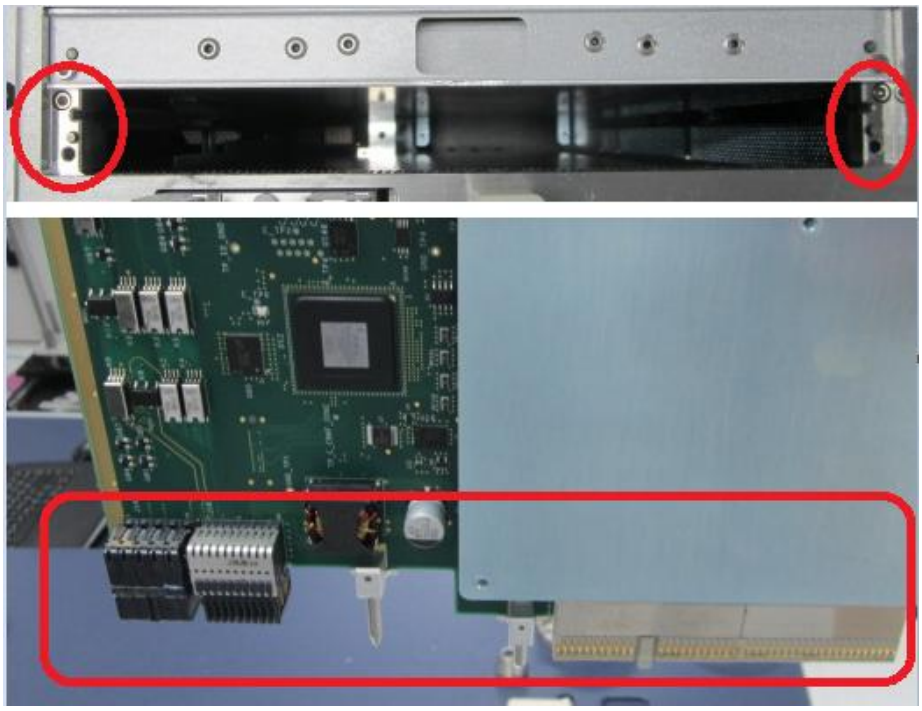




9.3 ระหว่างยกบอร์ดขึ้นมาให้จับบริเวณขอบบนของบอร์ด **(Interposer)** ด้วยมือทั้งสองข้าง **ห้าม**ไปจับที่อุปกรณ์บนบอร์ด



9.4 (การใส่)ให้สังเกต**ตำแหน่งไกด์ของSlot+รูไกด์ของบอร์ด** และ **Connector+ไกด์ด้านล่าง**ของบอร์ด

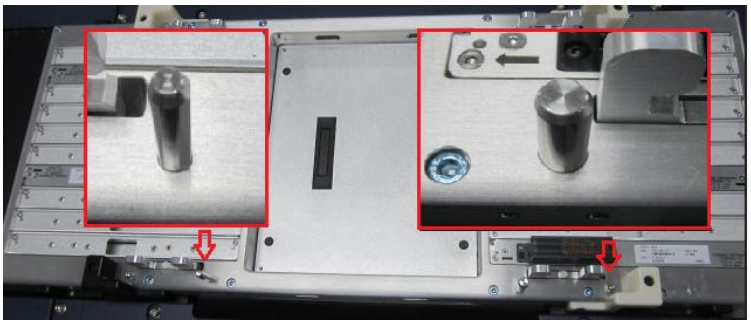


NOTE: The fans used for cooling the test head are controlled by circuitry that increases or decreases fan speed to meet cooling demand. When powering up the tester the fans will run at full speed. Once the controller turns on, the fan speed is reduced. Fan speed during tester operation, may be highly variable depending on the instruments in the test head and the temperature of the environment the tester is run in.

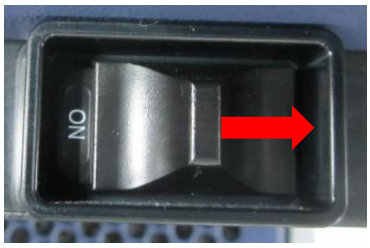
10. การใส่-ถอด Load board

10.1 (การใส่)ก่อนใส่ต้องเช็คก่อนว่า**DUT FIXTURE**และสวิตช์อยู่ในตำแหน่ง '**UNLATCH** (slidไปทางซ้ายให้สุด)' + '**OFF**'

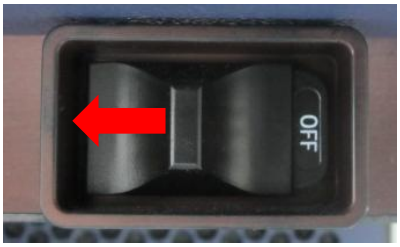
10.2 ใส่Load board ลงไปโดยให้สังเกตุตำแหน่งไกด์(เล็ก-ใหญ่) เป็นหลัก Load boardจะต้องลงพอดี



10.3 ถ้าลงทั้ง4มุมแล้ว ให้เปลียนตำแหน่ง **DUT FIXTURE** เป็น '**LATCH** (slidไปทางขวาให้สุด)' และเปลี่ยนสวิตช์เป็น '**ON**' มอเตอร์จะดึง Load board เพื่อล๊อกและจะต้องรอจนมอเตอร์หยุดทำงาน



10.4 (การถอด)เปลี่ยนตำแหน่งสวิตช์เป็น '**OFF**' และเลื่อนตัวล๊อกเป็น '**UNLATCH'** มอเตอร์จะปล่อย Load board เพื่อล๊อกและจะต้องรอจนมอเตอร์หยุดทำงาน



10.5 จากนั้นก็ให้ยก Load board ออก